HOPES TO EXPAND FUEL RANGE WIDELY

New Carburetor Principle Aimed at Use of the Heavier Oils Up to High Boiling Point.

HUGE WASTE AT PRESENT

Inventor Believes Fluid Should Be Nebulized Rather Than Atomized —Chicago Tests Succeed.

In spite of plans and undoubted intentions to build more automobiles in the present year than ever before, it is not denied that the motor industry, as well as the motor car owner, faces a most serious problem and possible setback in the soaring price of gasoline. It is predicted that "gas" will reach 40 cents or more a gallon in the coming Summer. What this means to the pleasure car maker and owner, as well as to those interested from any viewpoint in motor trucks, may readily be appreciated. What is not so generally understood, however, is the enormous waste of natural resources which the country suffers in the failure to use a wider range of the petroleum products as fuel for internal combustion engines. The figures are startling, to say the least.

Last year 220,000,000 barrels of crude oil were produced in this country. From it was produced roughly 10,600,000 barrels of motor gasoline. Less than 5 per cent. of the total crude oil was, therefore, available for motor purposes. After deducting one-half of the total supply of crude oil for all other purposes, including fuel oils, lubricating oils, and residum, about ten and one-half times the present amount, or 110,000,000 barrels, would remain, which could be used in internal combustion engines, provided the proper carburetor, or instrument for feeding it, were devised.

This would mean an addition to the motor fuel supply of 4,000,000,000 gallons, or an addition to the power supply of the country of some 48,000,000,000 horse-power hours, annually. The automobile industry, now so seriously jeopardized by the question of fuel shortage and price, could be carried to nearly eleven times its present extent. In terms of money, the failure to utilize as motor fuel the 100,000,000 barrels of crude oil, much of which is now sold at 3 cents a gallon and less, means an annual loss to the oil producers, at the present price of gasoline, 16 cents a gallon wholesale, of $500,000,000. Even if the throwing on the market of this huge supply of fuel brought down the price per gallon to, say, 9 cents, which was the gasoline price a year ago, the difference between this 9 cents and the 3 cents now obtained for the material not utilized as fuel would be about $240,000,000 annually.

The enormous quantities of fuel which would be furnished by the petroleum bases, if those oils up to, say, 600 degrees Fahrenheit, boiling point, could be utilized, are thus apparent. Experts are not wanting, who assert, conversely, that the petroleum bases are at present the cheapest source of motor fuel. Their argument runs somewhat as follows: The earth is now practically an ash pile in the sense that little remains uncombined which is of a heat or power giving nature. The products or compounds available for use in an internal combustion engine must be those which will produce high temperatures when combined with the oxygen of the atmosphere. Little remains of the earth's total bulk which is not already combined and burned up. Thus the free hydrogen is combined with sea water, forming the oceans. The few metals, such as gold, copper, silver and iron, not already in combination with oxygen are not available for use in an engine, since for this use one must have a material whose product of combustion is also gaseous.

This practically limits the available sources of motor fuel to carbon, or coal; hydrocarbon, the oil supply, and carbohydrates, such as cellulose, woody fibre and starch, from which alcohols can be derived. The chemists have no other choice; in order to produce a really new fuel they must make use of some unknown thing.

According to Henry M. Reichenbach, a chemist and engineer of Rochester, N. Y., who has had a long experience in invention, including some of the basic work on the photographic film, the carbohydrates are practically eliminated from the fuel field by reason of the expense attendant upon their production. The practical test, he believes, is one in which separation only, and not combination and then subsequent separation, is required to make it available for the internal combustion engine. Mr. Reichenbach believes that the solution of the problem lies in the use of the petroleum products to a wider extent than has hitherto been possible, and he believes also that he has devised a means to bring this about.

For eight years the inventor has been at work on a carbureter for the utilization of the heavier oils, and he has demonstrated that his perfected device shows thorough combustion with fuels ranging from low boiling points far up the scale, through kerosene and probably at least up to 600 degrees. The principle of this carburetor is to break up the fuel more completely than is now done, to nebulize rather than to atomize it; in other words, to bring the charge into the cylinder in the form of a fog rather than in the form of a spray. The theory behind the practice in this device is that if the particles of oil are propelled with a sufficient velocity through orifices the minute particles become charged in a like direction electrically and repel each other instead of coalescing. This is on the principle of the suspension of the clouds, and is based on the well-established tendency of particles, each positive or each negative, electrically to repel each other. In the carburetor it may be brought about by the exceedingly minute subdivision of the fluid, as contrasted with the coarser spray now utilized—by atomizing—which is easily condensible.

Mr. Reichenbach's carburetor may be briefly described as a device to use the suction of the motor to subdivide the fuel as far as possible. It is not a suction feed device, being supplied by gravity, as in many familiar carburetors. But by placing the throttle behind, rather than in front of the jets, the suction of the motor acts as the first powerful medium for breaking down the fluid character of the fuel.

In general plan the instrument is described as follows:

The entering air first passes through the throttle, thence by a gate through an opening, the area of which varies with the volume which the throttle permits to enter. At this point the fuel is mixed with it. The movement of the gate against the force of the calibrated spring tending to close it, also changes the rate of fuel supply by means of a calibrated valve. This change of the rate of supply of the fuel is so fixed that it gives maximum power and economy at all speeds. A little consideration will show that changes in barometer pressure which change the "sucking action" of the motor are automatically compensated for. Hence, when once adjusted in the factory, no other adjustment is ever necessary. This overcomes one of the most annoying troubles of other carburetors, i. e., the necessity of adjustment to compensate for changed weather conditions.

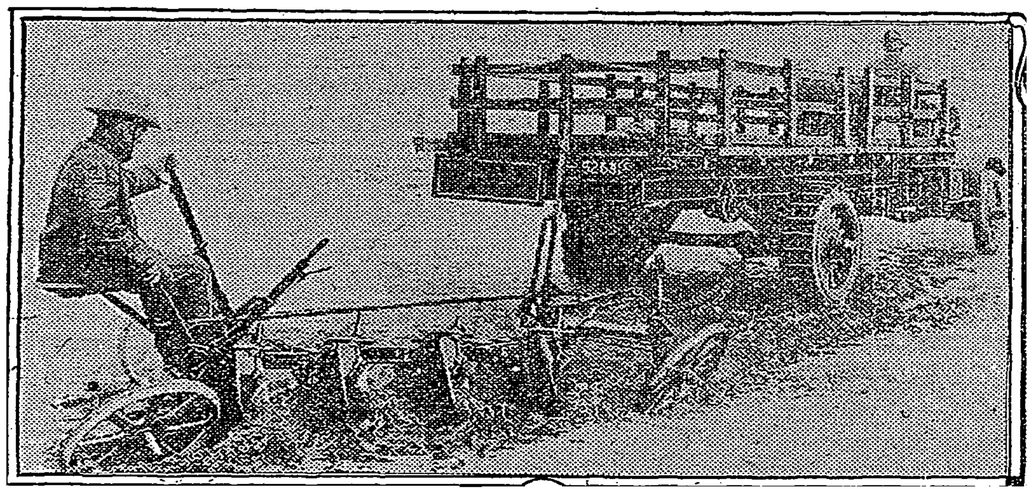
The device has been subjected to a test in the laboratories of the Armour Institute of Technology at Chicago, and Mr. Reichenbach received letters from G. F. Gebhardt and Arthur J. Frith, mechanical engineers connected with the laboratories, concerning the behavior of the carburetor. Mr. Gebhardt's letter runs:

"This is to certify that extensive experiments were conducted in our laboratories with a Reichenbach carburetor under various conditions of operation. The experiments were made under the direct supervision of Mr. Reichenbach. Although the writer was not personally engaged in the testing, he was present during a number of the runs, and is of the opinion that the device is an excellent one, and designed along correct principles. This carburetor, attached to a four-cylinder Rutenber motor, operated smoothly and economically for all loads within the capacity of the motor. One of the important features of this device is its adaptability to kerosene of low density. During one test, using kerosene of 50 degrees gravity, the speed of the engine was readily varied from 175 to 1,500 revolutions per minute; ignition was perfect and examination of the spark plugs after the test showed no soot deposit."

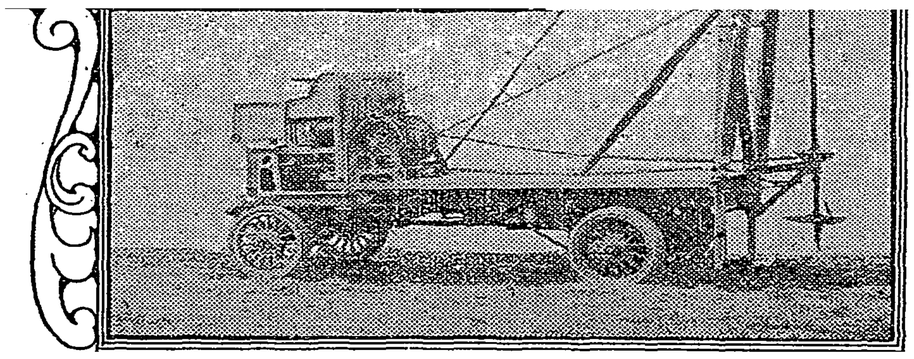
The letter from Mr. Frith is as follows:

"I take pleasure in stating that I witnessed the experiments made on a carburetor devised by Mr. Reichenbach, as attached to an automobile engine in the laboratory of the Armour Institute, and am impressed with its capability to give a uniform mixture to the engine throughout the entire range of speed and power that was tried on this machine. This was done without an adjustment or jockeying of the carburetor, and is the valuable feature of carburation that is so imperfectly attained in ordinary practice. Mr. Reichenbach's carburetor method of vaporizing at his throttle under a high vacuum is, in my experience, entirely novel, is certainly correct, and is, I think, the cause of the excellent results attained. He has also arranged it so that it should not be affected by changes of atmospheric pressure, and I know of no other device that even automatically attempts to handle this troublesome condition. I think the device has great possibilities."

MOTOR TRUCKS HAVE MANY USES



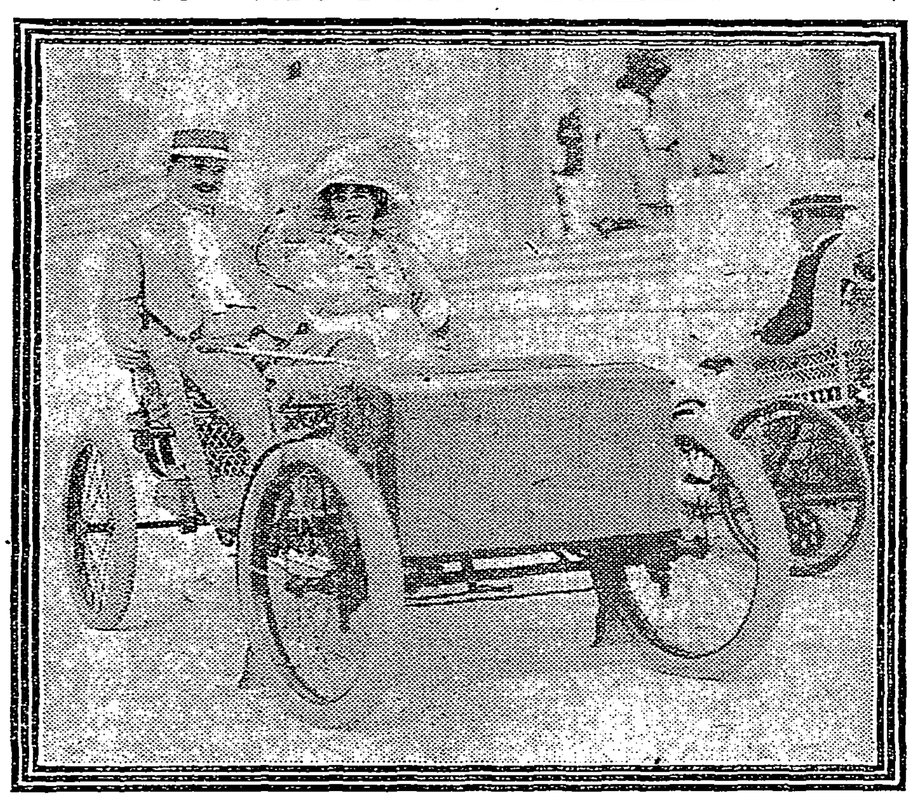
THREE TON TRUCK USED FOR PLOWING



TRUCK EQUIPPED WITH POWER WINCH FOR AUTOMATIC SETTING OF POLES. THIS TRUCK, IT IS REPORTED, IS DOING THE WORK OF SIX MEN.

The farmer has attached his plow to a Kissel Kar. The Mack truck below has seen service in the Middle West.

A "MOTOR ROLLER."



Mr. and Mrs. Edward B. McLean, of Washington, D. C., have introduced this novel form of locomotion at Palm Beach The machine is for their exclusive use and is said to cost $100 a week.
(Photo by Underwood and Underwood.)

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